Data-Analytic Thinking

Provost & Fawcett, Data Science for Business, chapters 1–2

AI in Business · Week 2

Today's plan

1

Why data-analytic thinking?

Chapter 1: cases, evidence, data as an asset

2

From business problems to data tasks

Chapter 2: nine task types, supervised vs. unsupervised

3

The data mining process

Chapter 2: CRISP-DM, lessons per stage, related techniques

1. Why data-analytic thinking?

Chapter 1, Provost & Fawcet

Walmart and Hurricane Frances: find the non-obvious

7× normal sales of strawberry Pop-Tarts in the days before a hurricane

#1 pre-hurricane top seller: beer

Obvious
People buy water and flashlights. No data science needed.

Valuable
Mine data from earlier storms (Hurricane Charley) for unusual local demand.

Actionable
Ship the right stock to stores before landfall.

Source: Provost & Fawcett ch. 1, citing the New York Times (Hays, 2004).

MegaTelCo: the running example of the book

20%

of mobile customers leave when their contract expires. Winning new customers is far more expensive than keeping existing ones.

The task

Marketing has designed a retention offer and has a fixed budget. Which customers should receive it before their contract ends?

Think about it

- What data would you use?
- Those most likely to leave? The most valuable?
- Or those an offer can actually change?

We return to this problem all semester. Today's Python lab uses a real telecom churn dataset.

Decisions, data science and data engineering

Data-driven decision-making (DDD)
Decisions based on data analysis, not intuition alone; by people or automated

Data science
Principles and data-mining techniques for extracting knowledge from data

Data engineering and processing
Databases, pipelines, "big data" technologies: support, not science

Evidence

4–6%

One standard deviation more data-driven firms -> higher productivity

1–3%

one standard deviation more use of big-data technologies → higher productivity

Sources cited in ch. 1: Brynjolfsson, Hitt & Kim (2011); Tambe (2012).

Two kinds of decisions where data pays off



Type 1: Discoveries

One-off decisions that need a new insight hidden in the data.

- Walmart: what will sell before a hurricane
- Target: predicting which shoppers are expecting a baby



Type 2: Repeated at scale

The same decision made millions of times. Even a small accuracy gain adds up.

- MegaTelCo: who gets a retention offer
- Credit scoring, fraud detection, online ads, recommendations

The Target case also raised ethical concerns about privacy: a theme for week 13.

Data is an asset, and sometimes you pay to get it: Signet Bank Case

~1990

The idea

Model profitability, not only default risk, and offer each customer different terms

Problem

No data

No bank had data on how customers react to terms never offered

Investment

Random offers

Signet Bank tested terms at random. Charge-offs rose from 2.9% to almost 6%

Payoff

Capital One

Spun off and became a top card issuer: 45,000 tests a year by 2000



Fairbanks & Morris

Lesson: the losses were the cost of acquiring data. Data and data-science talent are complementary assets: one is little use without the other.

Four fundamental concepts we will use all semester



Follow a process

Extracting knowledge from data has well-defined stages (CRISP-DM).

*Cross-Industry Standard Process for Data Mining



Find informative attributes

Which customer characteristics tell us something about churn, and how much?



Beware of overfitting

Look hard enough and you will find a pattern that does not hold on new data.



Evaluate in context

Is the result better than chance, or a smart default? How will it be used?

"Chemistry is not about test tubes": tools change every few years; these principles do not.

Activity 1: Would you invest?

Scenario 1

A fintech start-up in Tirana says its unique data on customers' mobile payments will predict who is about to switch banks. On that basis, it asks for a much higher valuation.

Discuss a set of questions that would test whether the claim makes sense. Use the four concepts from the previous slide.

From business problems to data tasks

Chapter 2, Provost & Fawcet

Business problems break down into standard tasks

Classification
Who will respond to the offer?

Regression
How much will this customer use the service?

Similarity matching
Which firms look like our best clients?

Clustering
Do our customers form natural segments?

Co-occurrence
Which products are bought together?

Profiling (Behavior)
What is normal card use for this customer? (fraud detection)

Link prediction
You and Karen share 10 friends: connect?

Data reduction
Turn viewing history into a few taste factors

Causal modeling
Did the ad cause the purchase?

Supervised Unsupervised Either

Supervised or unsupervised? It depends on the target

"Do our customers fall into natural groups?"

No target → unsupervised

Groups may or may not be useful for any decision

"Which customers are likely to cancel when their contract ends?"

Target: cancelled yes/no → supervised

Groups formed for a purpose: acting on churn

Classification

categorical target: will she churn?

Class probability

how likely is she to churn? (still classification)

Regression

numeric target: how much will she spend?

The target must exist in the data. Getting labels often costs money, and is a data-science investment.

Activity 2: Name the task

1. Which loan applicants will default?
2. How many units will each store sell next week?
3. Which supermarket products are bought together?
4. Is this card payment unusual for this customer?
5. Did the loyalty card make customers spend more?
6. Which companies resemble our 100 best clients?
7. Can our 50 branches be grouped into a few types?
8. Which contacts should LinkedIn suggest to you?

Building a model is not the same as using it

Data mining



Use



Keep the two apart, but let the way results will be used shape how you build the model.

3. The data mining process

Chapter 2: CRISP-DM

CRISP-DM: iteration is the rule, not the exception



CRISP-DM: Cross Industry Standard Process for Data Mining (Shearer, 2000).

Why a cycle?

- Business problems rarely arrive as clean data-mining problems
- Each pass teaches you more about the problem and the data
- Evaluation often sends you back to business understanding
- A first pass that does not solve the problem is not a failure

One lesson from each stage

Business understanding	Think through the use scenario first: what exactly will we do with the results?
Data understanding	Card fraud has reliable labels; Medicare fraud does not. Same problem name, different task.
Data preparation	Beware leakage: always leads to awesome model outcomes (better accuracy).
Modeling	Where the algorithms live. The next chapters of the book.
Evaluation	A model 99% accurate in the lab can still raise too many false alarms to pay off.
Deployment	"Your model is not what the data scientists design, it's what the engineers build."

Data science is managed like R&D, not software



Software project

- Clear requirements and milestones
- Success is unambiguous
- Measured by code shipped, tickets closed



Analytics project

- Exploratory; outcomes are uncertain
- Invest in pilots and throw-away prototypes
- Needs people who frame problems, prototype fast, design experiments

For our course project: frame the problem and test a prototype first; build the business case only once it works.

Four related questions, four different tools

Who are our most profitable customers?	**Database query (SQL)**
Is the difference from average customers real?	**Statistics: hypothesis test**
What characterises profitable customers?	**Data mining: find patterns**
Will this new customer be profitable? How much?	**Predictive model: classification / regression**

Choose statistics with care: US mean income in 2004 was over $60,000, the median only $44,389. Skewed data needs the median.

Five things to remember

1. The goal is better decisions, not models or big data
2. Data and analytical talent are strategic assets; invest in both
3. Break problems into standard tasks: classification, regression, clustering …
4. Supervised learning needs a target that exists in the data
5. Follow CRISP-DM, iterate, and evaluate in the business context